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Definitions

- **BFS (Builders FirstSource)** – Publicly traded building-products company (*NYSE: BLDR*) that supplies lumber, trusses, engineered wood, wall panels and building-components. BFS operates across many U.S. states and must design its products to comply with national codes and state or local amendments.

Introduction & Why Code Variance Matters

The U.S. does not have a single national building code. The **International Building Code (IBC)** and **International Residential Code (IRC)** provide model rules, but every state and local jurisdiction can adopt, amend or supplement those rules to reflect local hazards such as hurricanes in Florida, heavy snow in Colorado, or tornadoes in Kansas. **Builders FirstSource (BFS)** designs roof trusses, floor systems, wall panels and supplies dimensional lumber across the country. Because loads and design limits vary, BFS must adjust spans, select appropriate materials and design connections to meet each code. Failure to meet local requirements can lead to structural deficiencies or rework. This article compares national standards with several state examples to illustrate why understanding local code differences is critical for design and product selection.

National Codes & Standards (IRC, ASCE, IBC)

Model codes

- **International Building Code (IBC)** – Used for commercial structures. It references **ASCE/SEI 7** for environmental loads (snow, wind, seismic, rain and tornado) and sets risk categories for occupant safety. For example, the 2024 IBC adopts new reliability-targeted ground snow load maps and wind maps aligned with **ASCE 7-22** ¹. It also introduces tornado-load provisions for Risk Category III and IV buildings ².
- **International Residential Code (IRC)** – Applies to one- and two-family dwellings. The IRC contains prescriptive limits for structural elements. If a design exceeds those limits (e.g., very long spans or high loads), the IBC/ASCE design procedures must be used. The IRC limits residential truss designs to a maximum wind speed of **140 mph**, a ground snow load of **70 psf** and truss spans up to **36 ft** ³.
- **ASCE/SEI 7** – The load standard referenced by both IBC and IRC. It provides maps for wind speeds, ground snow loads and seismic accelerations. ASCE 7-22 adds risk-category-dependent snow loads and new tornado hazard maps ¹ ².

Deflection and moisture limits

- **Deflection limits** – Codes control the permissible bending of members. A MiTek engineering note summarises that roof trusses must limit live-load deflection to **L/240** (span/240) and total-load deflection to **L/180**, whereas floor trusses must limit live-load deflection to **L/360** and total-load deflection to **L/240** ⁴. Designers sometimes specify stiffer criteria such as L/480 for floors ⁵.
- **Moisture content** – Dimensional lumber used for framing should be kiln-dried to **≤ 19 % moisture content**. A wood-industry guide notes that **2×4 and 2×6 framing studs** should measure **19 % moisture or less**; subfloor panels such as plywood or OSB should measure between **10 % – 14 %** ⁶. Higher moisture can lead to warping and mold.

State / Climate / Seismic / Wind / Snow Zones

The table below summarises environmental loads for three regions (Florida, Colorado and Kansas/Midwest). States typically adopt the IBC/IRC but specify local design loads and amendments.

State/region	Climate hazards & code focus	Snow load (ground)	Design wind speed (Risk Cat II)	Notes
Florida (coastal/hurricane)	Hurricane winds, flood zones, high humidity. Since Hurricane Andrew, Florida has added stringent wind-borne debris and flood-proofing rules ⁷ . Wind-borne debris regions require impact-resistant windows or shutters for new construction ⁸ ; high-velocity hurricane zones (HVHZ) require windows that withstand 170–200 mph winds ⁹ .	Not typically controlling; ground snow load effectively zero in low elevations.	Wind zones: Zone 1 – $130 \leq V < 140$ mph; Zone 2 – $140 \leq V < 150$ mph; Zone 3 – $150 \leq V \leq 170$ mph; Zone 4 – $V > 170$ mph ¹⁰ . In wind-borne debris regions, ultimate design wind speed ≥ 130 mph or 140 mph ¹¹ .	Flood-proofing and elevation requirements; hurricane straps tie roofs to walls ¹² ; energy efficiency for hot climates ¹³ . Seismic hazard is low (SDC A).

State/region	Climate hazards & code focus	Snow load (ground)	Design wind speed (Risk Cat II)	Notes
Colorado (Front Range / mountainous)	Cold winters, heavy snow at high elevations, moderate winds and some seismic risk. Pikes Peak Regional Building Code (PPRBC) lists ground snow loads of 43 psf below 7,000 ft and 57 psf above 7,000 ft , with flat roof snow loads of 30 psf and 40 psf respectively ¹⁴ . Basic wind speeds (Risk Cat II) are 130 mph , increasing to 140 mph for Risk Category IV structures ¹⁵ .	43 psf (≤ 7 k ft) – 57 psf (> 7 k ft) ¹⁴	130 mph (Risk Cat II), 135 mph (RC III), 140 mph (RC IV) ¹⁵	Many Colorado jurisdictions require exposure category C for wind ¹⁵ . The Structural Engineers Association of Colorado recommends using seismic design category B or higher rather than SDC A because Colorado has documented earthquakes ¹⁶ ¹⁷ .
Kansas / Midwest (Hays, KS example)	Moderate snow loads, tornado risk (“Tornado Alley”), low seismic hazard. The City of Hays design criteria specify ground snow load = 25 psf and wind speed = 115 mph ¹⁸ ; frost depth 30 inches. Tornado-resistant design may be required for Risk Category III and IV buildings under ASCE 7-22 ² .	25 psf ¹⁸	115 mph ¹⁸	Tornado loads apply to RC III-IV buildings in tornado-prone regions ² . Seismic design category typically B or lower.

Snow and wind design differences

- **Florida:** Structural design is governed by wind. Structures near the coast must resist **130–170 mph** three-second gusts ¹⁰ and comply with impact-resistant glazing requirements ⁸. Snow loads are negligible in most of Florida.

- **Colorado:** The high snow loads (up to **57 psf**) require larger roof members and stronger trusses than in Florida or Kansas. Because the design wind speeds (130–140 mph) are similar to Florida’s Zone 1–2 values but snow load is much greater, truss manufacturers must adjust chord sizes and plate designs. Colorado jurisdictions also require exposure category C due to open terrain ¹⁵.
- **Kansas / Midwest:** Snow loads are lower (**25 psf**) and wind speeds moderate (**115 mph**), but tornado risk triggers checks for tornado loads on critical facilities ². Designers may need to provide safe rooms or storm shelters in some occupancies.

Seismic considerations

- **Colorado** – Historically experiences earthquakes; SEAC recommends using at least **Seismic Design Category B** for most buildings to ensure life safety ¹⁶ ¹⁷.
- **Florida and Kansas** – Generally low seismic hazard (SDC A or B). Structures may not require special seismic detailing.

Spec Adjustments by Region (span, deflection, moisture)

Member spans and loads

The capacity of lumber and engineered wood products is influenced by span length, spacing, grade and live load. A joist span table for **#2 grade lumber** with a live load of **40 psf** and dead load of **10 psf** shows that a **2×10** floor joist spaced 16 inches on center can span **about 14 ft (Southern Pine) to 15 ft 7 in (Douglas Fir)** ¹⁹. Under lighter 30 psf live loads, the same 2×10 can span 16 ft 2 in to 18 ft ²⁰. Snow loads or heavy finishes can reduce allowable spans.

Deflection limits

Codes limit deflection to avoid sagging and cracking. For roof trusses, the live-load deflection must not exceed **L/240** and total-load deflection **L/180** ⁴. Floor trusses must limit live-load deflection to **L/360** and total-load to **L/240** ⁴. When owners desire stiffer floors (e.g., for tile), designers may specify **L/480** ⁵. Regions with high snow loads (e.g., Colorado) often require stiffer deflection limits than hurricane-prone but low-snow regions like Florida.

Moisture considerations

Structural lumber should arrive at the job site with moisture content $\leq 19\%$ ⁶. Higher moisture content can lead to shrinkage and may reduce structural capacity. Colorado’s cold climate calls for additional measures to prevent condensation in roof cavities, whereas Florida’s humidity necessitates rot-resistant materials and ventilation to control moisture.

Cases: Florida vs Colorado vs Midwest (CoT example)

To illustrate how design loads drive product choices, consider a one-story residence using 2×10 floor joists and roof trusses. We step through the loads for each state:

1. Identify climate loads:

2. Florida's controlling hazard is wind. A house in Palm Beach County located in **Wind Zone 2** must be designed for **140 mph** ultimate wind speed ¹⁰ . Snow load is negligible.
3. Colorado's controlling hazard is snow. In Colorado Springs at 6,500 ft elevation, the **ground snow load is 43 psf** ¹⁴ and the basic wind speed is **130 mph** ¹⁵ .
4. Kansas (Hays) experiences moderate snow (25 psf) and **115 mph** winds ¹⁸ ; tornado loads may govern for essential facilities ² .

5. Select span and spacing:

6. Using the span table, a 2×10 joist at 16 inches on center can span ~14 ft under 40 psf live load ¹⁹ . For snow load design in Colorado, the effective uniform load may exceed 40 psf (e.g., 30 psf live + 43 psf snow), so the allowable span must be reduced. Designers might choose a **2×12** or engineered I-joist to achieve longer spans or adopt closer spacing.
7. In Florida, where wind governs, the floor loads remain standard. The same 2×10 span may be acceptable. However, roof trusses require increased uplift resistance (hurricane straps and ties) ²¹ and impact-rated sheathing.
8. In Kansas, designers must check tornado loads for Risk Category III/IV. Additional hold-down hardware or storm-shelter walls may be required.

9. Check deflection and connection requirements:

10. Roof trusses in Colorado must support heavy snow without excessive deflection. Designers may specify L/240 or stiffer criteria and include camber (pre-curvature) to counteract dead-load sag ⁴ .
11. Florida's high winds cause uplift; trusses are anchored with metal connectors (hurricane straps) ²¹ . Deflection under wind is less likely to govern, but shear walls and continuous load paths are critical.
12. Kansas roofs may be subject to tornado uplift; design must consider both straight-line winds and tornado loads for critical facilities ² .

13. Select materials and moisture controls:

14. In Florida's humid climate, pressure-treated lumber or engineered wood with decay resistance is preferred. All framing lumber should have moisture content $\leq 19\%$ ⁶ and must be protected from rain during construction.
15. Colorado's cold climate increases the risk of ice dams; roofs may require waterproof underlayment, ventilation, and thermal breaks.
16. Kansas's moderate climate requires standard moisture control; however, safe rooms may use reinforced concrete or panelized systems.

This case-by-case approach demonstrates how the same structural system must be adapted based on local loads and code amendments.

How BFS Products Map to Code Requirements

BFS supplies **roof and floor trusses, engineered wood products (EWP), wall panels, sheathing, insulation, and dimensional lumber**. The company's design services use computer-aided tools to engineer components that meet local codes. Examples of how BFS products align with codes:

- **Roof trusses** – BFS trusses are engineered using ANSI/TPI 1. Designs must respect the IRC span and load limits; for residences in Florida, trusses must withstand 130–170 mph wind pressures and include uplift hardware. In snowy Colorado, trusses use larger chord sizes and additional web members to support 30–40 psf roof snow loads ¹⁴ while meeting deflection limits (L/240 live, L/180 total) ⁴. For projects outside the IRC limits (e.g., spans > 36 ft or snow loads > 70 psf), BFS engineers follow IBC/ASCE 7 procedures and may recommend steel or heavy timber.
- **Floor systems** – BFS offers I-joists and open-web trusses that can span further than sawn lumber. When designing for Colorado's snow loads, BFS may increase member depth or select EWP with higher modulus of elasticity. In Florida or Kansas, standard I-joists may suffice. Deflection criteria (L/360 or L/480) are incorporated into BFS software to ensure floors feel stiff.
- **Wall panels and sheathing** – Factory-built wall panels are sheathed with OSB or plywood. In Florida's high-wind zones, panels must be nailed per high-wind schedules and may include hold-down straps. Colorado projects emphasise shear capacity for heavy snow; Kansas projects may include tornado-rated panels for safe rooms.
- **Lumber products** – BFS supplies kiln-dried structural lumber with moisture content below 19 % ⁶. This helps control shrinkage and complies with building code requirements. For outdoor or high-humidity environments (Florida), pressure-treated or rot-resistant species are offered.

Likely User Questions

- **What is the required snow load in Colorado and how does elevation affect it?** (Ground snow loads vary from **43 psf** below 7,000 ft to **57 psf** above 7,000 ft ¹⁴.)
- **Can a roof truss designed for Florida's 140 mph wind zone be used in Vermont or Colorado?** (Wind design may be sufficient but snow load differences require new designs; Colorado trusses must support heavy snow while Florida trusses emphasise uplift resistance.)
- **How does the required wind load differ between coastal and inland states?** (Florida coastal zones require 130–170 mph design speeds ¹⁰; inland Kansas uses 115 mph ¹⁸.)
- **Why does the IRC limit residential truss spans and wind speed to 36 ft and 140 mph ³?** (Beyond these limits, engineering analysis per IBC/ASCE 7 is required.)
- **What deflection limits apply to floor and roof trusses?** (L/360 for floor live load and L/240 for roof live load ⁴; designers may specify L/480 for stiffer floors ⁵.)
- **What moisture content is acceptable for framing lumber?** ($\leq 19\%$ moisture content for studs ⁶.)
- **How does tornado design differ from hurricane design?** (ASCE 7-22 introduces tornado wind speed maps; tornado loads apply to Risk Category III–IV buildings in tornado-prone regions ².)

Related Topics / Crosslinks

- **Truss Engineering and Design Tools** – See the BFS knowledge article on how BFS uses 3D modeling, ANSI/TPI 1 design and proprietary software to engineer trusses, I-joists and wall panels.
 - **Hurricane-Resistant Construction** – Learn about impact windows, shutters, continuous load paths and flood-proofing in BFS coastal projects.
 - **Snow Load Calculations** – An overview of ASCE 7 snow load equations and how BFS engineers determine roof snow loads at various elevations.
 - **Seismic vs Wind Loads** – Differences between seismic design category and wind design; how BFS structures are detailed for both hazards.
 - **Moisture and Wood Durability** – Best practices for storing and installing wood products in humid vs dry climates.
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1 2 2024 IBC Significant Structural Changes Part 6-Loads

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